CSC 473 - Automata, Grammars and Languages

MonWed 5:00PM - 6:15PM Haury Anthro Bldg, Rm 129

Course Description

This course is an introduction to the fundamental models of computation used throughout computer science: finite automata, pushdown automata, and Turing machines. The hierarchical relationships among these models, their relative power and limitations, and their variants are studied. Student skills are developed in understanding and using rigorous definition and proof to attack precisely formulated questions about computability and computation. The course aims at improving students' written communication skills and serves as the program's writing emphasis course.

Catalog Description: Introduction to models of computation (finite automata, pushdown automata, Turing machines), representations of languages (regular expressions, context-free grammars), and the basic hierarchy of languages (regular, context-free, decidable, and undecidable languages).

Instructor and Contact Information

Instructor:

Eric Anson

Office: Gould/Simpson 809

Tel: 520 621-2675

Email: eanson@arizona.edu

TAs:

Harshita Narnoli	harshitanarnoli@arizona.edu
Mayank Singh	mayanks43@arizona.edu

Office Hours

(Exact office hours subject to change, consult piazza for up-to-date information.)

Additional hours may be available by appointment.

Eric Anson

Mon, 1:00-2:30, GS 809

Tue, 3:30 – 5:00, GS 809

TAs Office Hours

TBA

Websites

Piazza: https://piazza.com/arizona/fall2025/csc473/home

D2L will be used for distributing grades, posting videos, and quizzes.

GradeScope will be used for returning exams and quizzes.

Course Format and Teaching Methods

There will be two lectures a week. Lectures will involve work on the whiteboard and input from students. Lectures will augment, enhance, and reiterate material from the textbook.

Course Objectives

1. Learn about the theory that underlies modern computing.
2. Be able to explain and prove some of the major results of this theory.
3. Construct, modify, and make statements about DFAs, PDAs, Turing Machines and the languages they describe.

Expected Learning Outcomes

Students will be able to:

1. Discuss and define regular, context free, and recursively enumerated languages and for a given language define an automata, regular expression, context free grammar, or Turing machine that represents that language.
2. Use the pumping lemma to prove a given language is not a regular language.
3. Define the classes P and NP, state their significance, and explain why the halting problem has no solution.

Makeup Policy for Students Who Register Late

Students who register late will not be able to make up the missed work.

Course Communications

The primary path for outside-lecture questions will be Piazza, though grading matters will go through GradeScope. Written assignments will be posted on D2L. The instructor and TAs can also be reached through email.

Required Texts or Readings

Sipser, Michael; *Introduction to the Theory of Computation*, 3rd Edition; (Cengage Learning)
ISBN-13: 978-1133187790

ISBN-10: 113318779X

Note this book is required. Lectures will closely follow the text and assignments may come directly from problems in the book. The book should be available for electronic rental through the bookstore (there should be a link on D2L). If you buy a copy somewhere else be aware there is an "international" soft cover version of the book (it's actually not supposed to be sold in the USA) that does <u>not</u> have the same problem set as the book we will be using. Since the assignments will come from the book, you want to be sure you get the correct copy.

Assignments and Examinations: Schedule/Due Dates

<u>Assignments</u>

Written assignments will be given weekly on most weeks. Assignments will usually be problems from the textbook. Sometimes answers to these problems will be available. For those cases students are expected to check the correctness of their work themselves, asking questions about answers they don't understand. Each assignment will have the same overall weight toward the final grade, and the assignment with the lowest score will be dropped.

Late assignments will only be accepted by arrangement with instructor or TAs.

Midterms

There will be two midterm exams. They will be held during the regular lecture times on Wed, Oct 8, and Wed, Nov 5.

Final

The final is scheduled for Thu, Dec 18, 1:00pm - 3:00pm

Final Examination

There will be a cumulative final given.
The final is scheduled for
Thu, Dec 18, 1:00 – 3:00 PM

https://registrar.arizona.edu/faculty-staff-resources/room-class-scheduling/schedule-classes/final-exams

Grading Scale and Policies

By department policy, the final exam is *required*.

I use the common 90-80-70-60 grading scale. It's possible that final grade cutoffs will be lowered a little (from 90% to 88.5% for the bottom of the 'A' range, for example) but they will never be raised. I make such determinations only at the end of the term, after the final exam has been graded.

The grades will be weighted as follows:

Assignments	=	20%
Participation	=	6%
Midterm Exams (22% each)	=	44%
Comprehensive Final Exam	=	30%
Total	=	100%

Assignments will typically be graded within a week of when they are due. Exams will be graded at most 10 days after they are given. If exceptions must be made occasionally, staff will inform students about the delay and the reason for it.

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at https://catalog.arizona.edu/policy/courses-credit/grading/grading-system.

Dispute of Grade Policy

If you believe an assignment or exam has been graded incorrectly, you must bring it up within two weeks after the item is returned to the class.

Scheduled Topic and Activities

This is a list of the topics we will cover. The dates may change slightly. Every semester the pace moves a little differently based on student questions, etc.

Week 1	Aug 25, 27	Introduction, Review, DFAs (Chap 0, 1)
Week 2	Sep 1, 3	Labor Day (holiday), DFAs & NFAs (Chap 1)
Week 3	Sep 8, 10	NFAs, Regular Expressions (Chap 1)
Week 4	Sep 15, 17	Regular Expressions, Pumping Lemma (Chap 1)
Week 5	Sep 22, 24	Pumping Lemma (Chap 1)
Week 6	Sep 29, Oct 1	Context Free Grammars (Chap 2)
Week 7	Oct 6, 8	PDAs (Chap 2), Exam 1 (Chap 1)
Week 8	Oct 13, 15	PDAs, Pumping Lemma for CFLs (Chap 2)
Week 9	Oct 20, 22	Pumping Lemma for CFLs (Chap 2)
Week 10	Oct 27, 29	Pumping Lemma for CFLs, Turing Machines (Chap 2, 3)
Week 11	Nov 3, 5	Turing Machines, Exam 2 (Chap 2, 3)
Week 12	Nov 10, 12	Variations of Turing Machines (Chap 3)
Week 13	Nov 17, 19	Church-Turing Thesis, Decidability (Chap 3, 4)
Week 14	Nov 24, 26	Undecidability, Reduction (Chap 4, 5)
Week 15	Dec 1, 3	Reduction, Complexity, the class P (Chap 5, 7)
Week 16	Dec 8, 10	NP and NP-Complete (Chap 7)

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Students are asked to refrain from disruptive conversations with people sitting around them during lecture. Students observed engaging in disruptive activity will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave lecture or discussion and may be reported to the Dean of Students.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): https://cirt.arizona.edu/case-emergency/overview

Also watch the video available at

https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/app/me/ledetail;spf-url=common%2Flearningeventdetail%2Fcrtfy000000000003841

University-wide Policies link

Links to the following UA policies are provided here: https://catalog.arizona.edu/syllabus-policies

- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy Class Recordings
- Additional Resources

- Preferred Names and Pronouns

Department-wide Syllabus Policies and Resources link

Links to the following departmental syllabus policies and resources are provided here, https://www.cs.arizona.edu/cs-course-syllabus-policies :

- Department Code of Conduct
- Illnesses and Emergencies
- Obtaining Help
- Confidentiality of Student Records
- Land Acknowledgement Statement

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.